



Estimation and prediction of the air–water interfacial tension in conventional and peptide surface-active agents by random Forest regression

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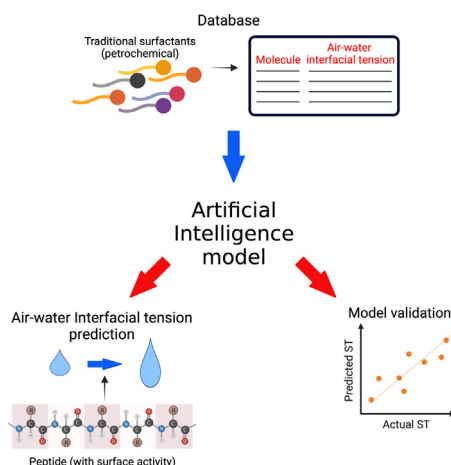
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HIGHLIGHTS

- A Random Forest model evaluated via a fivefold cross-validation was put forward.
- The model was used for estimating the air–water interfacial tension of various surface-active species.
- A database with the air–water interfacial tension of 691 surfactants was used for training the model.
- Structure-performance relationships were obtained for surface active molecules.
- Sequences of amino acids are proposed for screening peptides with high interfacial activity.

GRAPHICAL ABSTRACT



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ABSTRACT

The surface tension at the critical micelle concentration (ST_{CMC}) is an important descriptor for surfactants in applications including cosmetics, pharmaceuticals and food. A predictive ST_{CMC} Random Forest model was trained with 691 conventional surfactants and 9 amino acids. The model was evaluated by fivefold cross-validation, and the prediction power was tested for peptides by direct comparison with the experimental ST_{CMC} values found in the literature or measured in this work. Predictions were also conducted for short peptide permutations. The estimated ST_{CMC} approached the experimental values as the peptide length decreased, suggesting a strong influence of secondary structures for longer sequences where the developed algorithm fails to make robust predictions. Concerning the short peptide permutations, the model estimated lower ST_{CMC} values as the carbon number increased within the hydrophobic portions. This highlights the importance of the hydrophobic amino acids leucine, isoleucine, and phenylalanine in peptides with attractive surfactant properties.

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1. Introduction

Surfactants are chemical species that adsorb at interfaces and surfaces, reducing the interfacial free energy or interfacial tension due to the amphiphilic nature of the molecules (hydrophilic and lipophilic) (Rosen, 2004). These characteristics make surfactants suitable for formulating various products of commercial interest, including detergents, lubricants, emulsifiers, de-emulsifiers, inks, adhesives, cosmetics, drug delivery systems, pharmaceuticals, corrosion inhibitors, foods, coatings, oil recovery preparations, and liquid explosives, (Shaban et al., 2020; Pradilla et al., 2015; Das et al., 2020) only to mention a few. Designing and selecting surfactants involve critically evaluating physicochemical properties such as the critical micelle concentration (CMC), the cloud point (CP), and the surface tension (ST). An extensive knowledge of these properties is critical for estimating the surfactant capacity required to perform the desired function within the system of interest effectively. For example, the evaluation of the CMC is essential because, above this concentration, an abrupt change is produced in other surfactant properties such as the ST, the electrical conductivity, and the refractive index (Khan and Shah, 2008). Therefore, detergents must be formulated above the CMC such that they can solubilize lipophilic dirt by encapsulating it into micelles (Cheng et al., 2020). However, surfactants can be used below the CMC to precipitate and recover proteins in biological applications (Wong et al., 2018). The formation of surfactant micelles in water can be also exploited in catalysis as they can encapsulate the reagents and produce a controlled environment for chemical reactions (Maiti et al., 2018). Another important surfactant property is the CP or the temperature above which the surfactant forms a separate phase in water (making the solution cloudy). The CP is essential for evaluating product stability because if the storage temperature exceeds it, the product will separate over time into two phases (Al-Sabagh et al., 2011). In water remediation applications, an increment in the temperature up to the non-ionic surfactant CP allows phase separation and consequently, the extraction of pollutants such as polycyclic aromatic hydrocarbons (PAHs) (Li and Chen, 2008). In household applications, the wetting behavior and the foaming capacity are significantly reduced at and above the CP; for this reason, the surfactant must have a cloud point larger than the temperature of the application (Cohen and Rosen, 1981; Chaisalee et al., 2003). Finally, the ST is a vital surfactant property that is directly related to the surfactant performance. In products for agriculture and personal care sectors such as herbicides and body lotions, the preferred ST values are those in the low range because this ensures ease of spreading on surfaces (Ivanova et al., 2012). In fire-fighting and detergents, good foaming properties of water solutions are attained by ST reduction (Sheng et al., 2019). In inkjet printing, tiny drops are produced by reducing the ST, and thus, high-resolution images can be printed into the paper (Swerin and Mira, 2014). Product formulation in food, personal care and agriculture usually involves emulsion preparation to ensure even dispersion of different components. These emulsions are produced by reducing oil–water ST aided by surfactants (Stebe et al., 2001).

Surfactant properties can be measured by different techniques and have been collected for decades. Although and their availability has facilitated enormously the selection of surfactants for different applications, this methodology presents some drawbacks. First, laboratory techniques for research require high precision equipment and high purity chemicals that involve significant investments in purchasing and maintenance (Muñoz Alcuria et al., 2021). Furthermore, experiments require trained people who invest time and effort and might be exposed to risks that are inherent to laboratory work (Selvaraj et al., 2018). This human factor makes experimental results susceptible to mistakes and bias. Predictive mathematical models have emerged to estimate surfac-

tant properties as an alternative to testing procedures to overcome these drawbacks. Estimation models save resources by circumventing experimental evaluations, help understand structure–performance relationships, and allow optimizing the design and selection of molecules, thereby maximizing the desired properties while reducing the adverse or undesirable effects. Some models have discovered and quantified very precisely structure–property–performance relationships, which has been valuable for predicting whether a set of molecules is useful for a particular application. For example, predictions that are popular are the IT reduction for smaller transversal areas of surfactant molecules, and the higher CPs as the number of polyoxyethylene groups in ethoxylated surfactants increases (Rosen, 2004; Sheng, 2011; Haddou et al., 2019). In terms of the surfactant properties optimization for product design, Mattei et al. (Mattei et al., 2014) proposed a methodology in which an overall objective function is optimized based on some constraints and supported by models that describe how product properties change by the presence of different excipients, including the surfactants in the formulation. This methodology is an example of how important and powerful prediction models are in the optimal design and selection of surfactants.

Generally speaking, surfactant properties can be estimated *in silico* by two different methodologies: molecular dynamics simulations (MD) and artificial intelligence algorithms (AI). MD has been used to estimate properties such as the change of the surfactant-free energy during insertion into oil–water interfaces (Aguilera-Segura et al., 2016), the size and shape of surfactant micelles (Faramarzi et al., 2017), the estimation of anionic surfactant foaming ability (Xu et al., 2020), and the effect of salt molecules on the interfacial tension reduction caused by non-ionic surfactants (Alonso et al., 2020). MD simulations are advantageous for two reasons. First, starting values of the evaluated property are unnecessary. Second, the effect of different system conditions such as pH, temperature, and ionic strength needs to be taken into account (Ricardo et al., 2021). However, the short time and space scales required for the simulations may provoke remarkable differences between the real and the predicted properties (Aguilera-Segura et al., 2016; Faramarzi et al., 2017). The limited computer power is eventually one of the reasons for these differences and for the long-lasting simulations, which prevent evaluating properties of a larger number of molecules, a situation that is desirable for an optimization design task. These MD constraints are not present in AI techniques, which instead rely on a wide variety of machine learning protocols ranging from simple equations to intricate Artificial Neural Network (ANN) models. Many authors have attempted to develop mathematical equations to predict surfactant properties based on the molecular structure–property relationships during the past two decades. However, their prediction capabilities are extremely limited (Huibers et al., 1996; Huibers et al., 1997; ling Chen et al., 2007; Setiawan et al., 2020; Wang et al., 2005). These mathematical equations are also known as quantitative structure–property relationships (QSPR), which use molecular structure features such as the number of functional groups or atoms in the molecule to fit equations that can compute properties. Different QSPR equations have been proposed to estimate surfactant properties, such as the CMC of nonionic ethoxylated surfactants (Huibers et al., 1996; Wang et al., 2005), cationic gemini surfactants (Setiawan et al., 2020), and anionic sulfate surfactants (Huibers et al., 1997). Also, the CP of nonionic ethoxylated surfactants (ling Chen et al., 2007).

QSPR equations yield accurate results when the type of surfactants is very restricted. For example, QSPR equations for the CMC estimation of nonionic ethoxylated or anionic sulfates surfactants (Huibers et al., 1996; Huibers et al., 1997). However, when the evaluated surfactants show very different molecular structures,

the QSPR methods fail to provide precise predictions. Alternatively, AI methods such as Random Forest Regression (RF) and ANN models have emerged as a possibility to produce more precise property predictions while avoiding restrictions on the type of surfactant evaluated. As for the QSPR approaches, these methods may also use structural information for the calculations. However, the final model consists of a mathematical equation in which the estimated property and the structural descriptors are interconnected using simple mathematical functions (Huibers et al., 1996; Huibers et al., 1997). On the one hand, RF models predict properties by learning databases using several decision trees as building blocks, which decide upon the descriptors that minimize the error of the predictions (Schonlau and Zou, 2020). On the other hand, ANNs simulate the behavior of human body neurons represented by nodes; thus, learning and predictions are produced by the result of the interaction between different nodes (Zou et al., 2008). The precision of RF and ANN has led to multiple studies in various fields, including the prediction of the activity of antimicrobial peptides (Andreu et al., 2015), the prediction of thermophysical properties of liquid mixtures (Baskaran et al., 2008), and the prediction of tensile properties of alloys (Merayo et al., 2020). The use of RF or ANNs for surfactant properties estimation is still in its infancy mainly because a large amount of data is necessary to generate reasonable predictions (Çolak, 2021; Goodfellow et al., 2016). Thus far, there is a lack of surfactant property databases with a large number of molecules (>1000). The largest number of surfactants in a property database is provided by the National Standard Reference Data System, which includes information related to the CMCs and experimental measurement methods for around 721 surfactants (Mukerjee and Mysels, 1971). However, the identification of surfactants in this database is tedious and imprecise because of the use of commercial names, weak molecular structure information, and lack of cross-correlation with other sources of information. Many surfactants in the literature are frequently named differently depending on the author or the publication; thus, establishing the molecular structure for mathematical modeling purposes becomes difficult. Despite these data limitations, some authors have used RF and ANNs to estimate surfactant properties considering 51 to 202 cases (Astray et al., 2013; McDonagh et al., 2020; Qin et al., 2021). All these studies have focused on assessing the CMC. Other relevant properties such as the CP, Krafft Point, HLB, and ST have not been predicted by ANNs or RF models.

Regarding the ST, today's prediction capacity relies mostly on QSPR equations. The focus of these QSPR studies is to predict the ST at the CMC (ST_{CMC}) because this is the maximum ST reduction attained by a surfactant when its concentration in an aqueous solution is increased. The ST_{CMC} of nonionic ethoxylated surfactants was fit by a QSPR equation considering 30 molecules and obtaining a correlation coefficient (r^2) of 0.99 (Wang et al., 2005). The exact property was estimated for a group of anionic surfactants by using 20 model molecules to generate a QSPR equation that yielded an r^2 of 0.99 (Wang et al., 2003). More recently, the ST_{CMC} was modeled for 70 sugar-based surfactants obtaining a standard error of 2.4 mN/m in a validation set (Gaudin et al., 2018). Although fitting the two first models resulted in a high r^2 (Wang et al., 2005; Wang et al., 2003), the authors failed to check the prediction capacity of the models with a validation set. In addition, because the QSPR models developed for ST_{CMC} estimation are very restrictive in the type of surfactant used, the scope of the models is limited to molecules with very similar characteristics to those used for training.

The lack of models that can predict the ST_{CMC} limits their applicability to conventional surfactants (i.e., derived from petroleum) and protein- and peptide-based biosurfactants for which the task is even more difficult, because of their high tendency to form secondary and tertiary structures that might alter significantly the final properties (Ricardo et al., 2021). This insufficient performance

data has prevented the full replacement of petroleum-derived surfactants with biosurfactants, which are thought to be more favorable in terms of biodegradability, toxicity, and environmental impact (Tao et al., 2019). The performance of peptides as biosurfactants has been evaluated through interfacial activity measurements in silico via MD simulations and mathematical series which allow predicting the amphipathicity of the molecules based on the magnitude of the hydrophobic vector, a feature directly related to their interfacial activity (Aguilera-Segura et al., 2016; García-Moreno et al., 2020; García-Moreno et al., 2020). These approaches fail to provide a direct measure of IT and consequently, only provide a set of candidates that possibly have the potential to show interfacial activity. Applying AI models by considering conventional surfactants for training and their subsequent use in peptide ST_{CMC} estimation may help elucidate the molecular-property relationships underlying interfacial activity (García-Moreno et al., 2020; García-Moreno et al., 2020). Short peptides should eventually be the starting point for this purpose due to their lack of secondary structures; however, adjustments and further analyses can be developed to widen the peptides size application range.

The present study aimed to use a RF model trained with the molecular structural information of conventional surfactants and nine (9) amino acids as a new methodology for estimating and predicting the ST_{CMC} of petrochemical compounds and peptides with surface activity. For the application of the RF model, a large database of surfactant molecular structures versus ST_{CMC} values was developed, which included different types of surfactants (e.g., non-ionic, and cationic). The ability of the model to provide precise ST_{CMC} predictions over conventional surfactants with other molecular structures was evaluated by a fivefold cross-validation. The correlation coefficient (r^2), mean average error (MAE), mean squared error (MSE), root mean square error (RMSE), index of agreement (d), and accuracy (a) were used as metrics to evaluate the model performance over the validation sets. Finally, the model was used to rank a permutation of peptides with less than six (6) residues according to their predicted ST_{CMC} values. An analysis of the best-ranked peptides was developed, and therefore peptides with the most suitable molecular characteristics to reduce the ST_{CMC} were proposed to be validated experimentally in future studies.

2. Materials and methods

2.1. Materials

The ST of six amino acids was experimentally evaluated in this study. L-lysine (K) monohydrochloride and L-histidine (H) were purchased from Merck KGaA (Darmstadt, Germany), L-arginine (R) from Scharlab S.L. (Barcelona, Spain), glycine (G) from ChemCruz (Dallas, TX, USA), DL-methionine (M) and DL-tryptophan (W) were obtained from Sigma-Aldrich (St. Louis, MO, USA). Type I purified water (resistivity > 18.2 MΩ·cm) was used to prepare all solutions. The peptides RD (sequence: RTLFVCRVGD and purity: >95 % by HPLC) and NL-19 (sequence: IYKTPPIKDFGGFNFSQIL and purity: 95.3 % by HPLC) were synthesized from GL Biochem Ltd. (Shanghai, China) and their masses were confirmed via mass spectrometry.

2.2. Data acquisition

The surfactant database was built by collecting different ST_{CMC} values that were obtained previously experimentally and reported in the literature. The ST_{CMC} was provided in most studies, while other reports showed the IT vs surfactant concentration plot. In the second case, the ST_{CMC} was identified as the point at which an increase in concentration showed no impact in the measured IT (Labena et al., 2020). The experimental ST_{CMC} values reported in this study involved different measurement techniques (i.e.,

Wilhelmy plate, du Nouy ring, and pendant drop). Only sources that allowed a precise identification and evaluation of the surfactants were considered. In other words, polymeric surfactants, surfactant mixtures, and surfactant solutions with buffers or salts were excluded from the database. The test temperature ranged from 20 °C to 30 °C where the ST_{CMC} remained constant. Surfactants with >400 atoms were also excluded because their tendencies to form secondary and tertiary structures that alter the ST_{CMC} . Optical and stereoisomers were not categorized separately.

The molecular structure was drawn in the PubMed Sketcher and exported as a SMILES code for further analysis (Center, 2021). The SMILES is a 2D representation of the molecules that allowed visually extracting some descriptors to train the RF model. Finally, the amino acids K, R, G, M, and H were added to the database and their ST was experimentally measured as a function of the concentration by the pendant drop technique with a tensiometer Attention Theta Optical (Biolin Scientific, Gothenburg, Sweden). The lowest value of the ST was taken instead of ST_{CMC} because the amino acids failed to show surface activity at the air–water interface, and therefore a CMC is not attainable in water solutions (Ricardo et al., 2021; Chandra et al., 2013). Other three amino acids were considered, and their lowest ST were extracted from other studies (Chandra et al., 2013; Infante and Moses, 1994).

2.3. Descriptors definition

According to the molecular characteristics of the assessed surfactants, 37 descriptors were manually included in the database (Table 1).

One of the main molecular characteristics of the surfactants is the presence of hydrophilic and hydrophobic moieties. In the case of peptides, the hydrophilic region is comprised by the backbone of the peptides, while the hydrophobic regions or motifs are composed by the carbon chain ramifications. For this reason, some functional groups were labeled as “int” and “ext” to help identifying those located within hydrophilic and hydrophobic portions of the molecule, respectively. The hydrophilic surfactant moiety was defined as the portion of the molecule that contains all the electronegative atoms. A carbon is considered as part of the hydrophilic moiety when is not located in the pathway between two electronegative atoms (i.e., N, O or S). For example, the descriptor LONGHIDR, which accounts for the total number of atoms in the hydrophilic moiety, has a value of 1 for the surfactant di-dodecyl dimethyl ammonium bromide which represents the nitrogen between carbons 12 and 15 (Fig. 1 (A)). The carbon atoms 13 and 14 were not counted in LONGHIDR because they are located at the tails of the molecule. For the surfactant 8–2–8 (Fig. 1 (B)), the descriptor LONGHIDR, which has a value of 26, considers all the atoms between the oxygen located at carbons 8 to 9 and the oxygen between carbons 28 to 31. The carbons 17, 18, 21 and 22 are excluded from LONGHIDR because they are at the tails of the molecule. For the peptide VTV, the descriptor LONGHIDR counts the 15 atoms between the nitrogen bonded to Carbon 4, and the oxygen bonded to Carbon 14, excluding the carbons 1, 2, 3, 8, 11, 12, and 13, since they are outside the hydrophilic moiety (Fig. 1 (C)). The descriptor CARBINT accounts for the carbons along the hydrophilic moiety but excludes those within the functional groups C=O and COOC because these have been already considered in the descrip-

Table 1
Names and description of the descriptors used for training the RF model.

Code	Descriptor	Description	Maximum value	Minimum value
A	CH	Molecule net charge	4	0
B	ANI	Number of rings	4	0
C	LONGCAD	Number of carbons of the largest carbon external chain	26	0
D	ATOMTOT	Total number of atoms	206	10
E	LONGHIDR	Number of atoms of the hydrophilic moiety	190	1
F	CARBEXT	Total number of carbons in the hydrophobic moiety	56	4
G	CARBINT	Total number of carbons in the hydrophilic moiety	126	0
H	SECCEXT	Number of sections in the hydrophobic moiety	12	1
I	SECCINT	Number of carbon sections in the hydrophilic moiety	63	0
J	CH2-INT	Number of CH ₂ groups in the hydrophilic moiety	126	0
K	CH-INT	Number of CH groups in the hydrophilic moiety	10	0
L	C-INT	Number of C groups in the hydrophilic moiety	2	0
M	BENZ-INT	Number of benzene groups in the hydrophilic moiety	2	0
N	BENZ-1S-INT	Number of benzene rings with one carbon replaced by a non-hydrogen atom	3	0
O	AR5-2S-INT	Number of 5-carbon aromatic rings with two of the carbons replaced by non-hydrogen atoms in the hydrophilic section	2	0
P	L5-1S-INT	Number of 5 carbon rings with one of them different to a carbon in the hydrophilic portion	2	0
Q	L5-2S-INT	Number of 5 carbon rings with two of them different to a carbon in the hydrophilic portion	2	0
R	L6-1S-INT	Number of 6 carbon rings with one of them different to a carbon in the hydrophilic portion	2	0
S	L6-2S-INT	Number of 6 carbon rings with two of them different to a carbon in the hydrophilic portion	2	0
T	CH2-EXT	Number of CH ₂ groups in the hydrophobic portion	51	0
U	CH-EXT	Number of CH groups in the hydrophobic portion	2	0
V	C-EXT	Number of C groups in the hydrophobic portion	2	0
W	BENZ-EXT	Number of benzene groups in the hydrophobic portion	2	0
X	C=C-EXT	Number of double bonds C=C in the hydrophobic portion	2	0
Y	COOC	Number of ester groups	4	0
Z	C=O	Number of C=O bonds different to ester	4	0
AA	—O—	Number of ether groups	63	0
BB	—S—	Number of Sulphur atoms with two carbon bonds	1	0
CC	N-TERT	Number of nitrogen atoms with 0 net charge and three carbon bonds	3	0
DD	NH-SEC	Number of nitrogen atoms with 0 net charge and two carbon bonds	6	0
EE	NH2-PRIM	Number of nitrogen atoms with + 1 net charge and one carbon bonds	2	0
FF	N + CUAT	Number of nitrogen atoms with + 1 net charge and four carbon bonds	4	0
GG	NH + TERT	Number of nitrogen atoms with + 1 net charge and three carbon bonds	2	0
HH	NH2 + SEC	Number of nitrogen atoms with + 1 net charge and two carbon bonds	3	0
II	NH3 + PRIM	Number of nitrogen atoms with + 1 net charge and one carbon bond	2	0
JJ	OH	Number of hydroxyl groups	12	0
KK	CH3	Number of CH ₃ groups	12	0

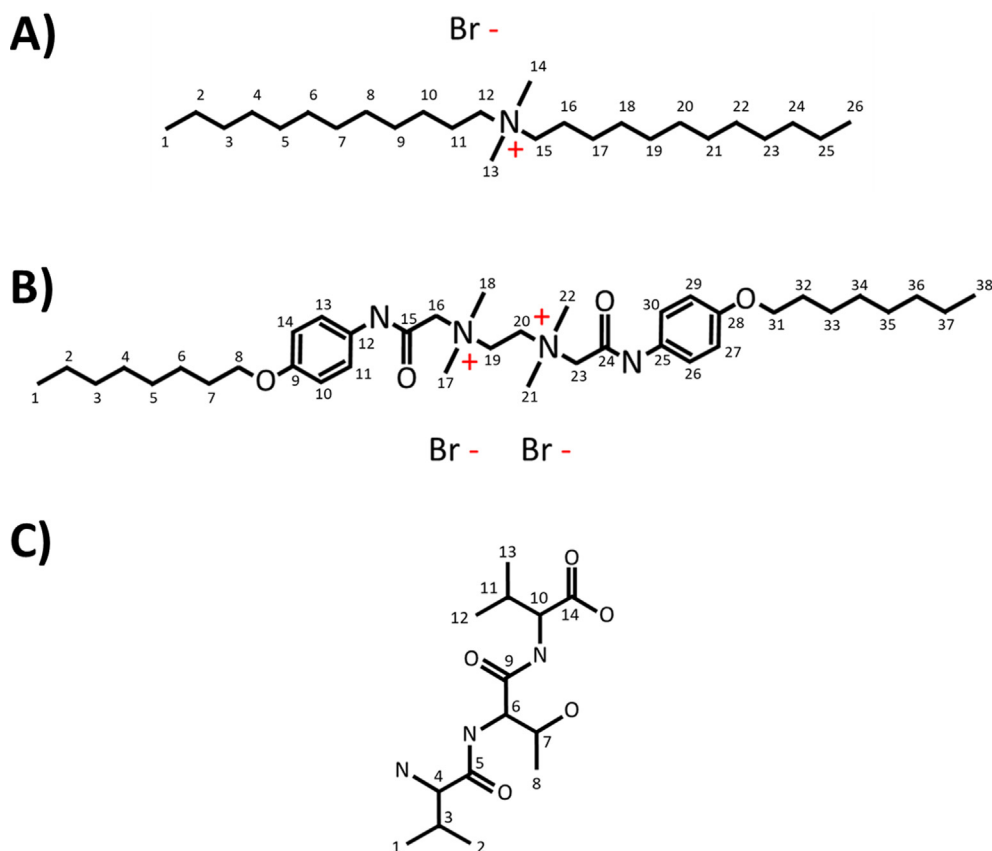


Fig. 1. Molecular structure of the surfactants di-dodecyl dimethyl ammonium bromide (A), 8-2-8 (B), and the peptide VTV, with all the enumerated carbons to show how the descriptors of the models were defined.

tors “C=O” and “COOC,” respectively. In the case of di-dodecyl dimethyl ammonium bromide (Fig. 1 (A)), no carbons are present along the hydrophilic moiety (i.e., CARBINT = 0), while in the 8-2-8 surfactant, the value of CARBINT is 16. Concerning the peptide VTV, the descriptor CARBINT has a value of 4 because it considers the carbons 4, 6, 7, and 10 (Fig. 1 (C)). Eventually, carbons or groups of carbons that belong to the hydrophilic moiety and are accounted in CARBINT, are separated by electronegative atoms, as can be seen in Fig. 1 (B). These groups are considered hydrophilic and accounted for by the descriptor SECCINT. Fig. 1 (A) shows no such moieties for the di-dodecyl dimethyl ammonium bromide, while Fig. 1 (B) allows identifying five inner sections of the 8-2-8 molecule. The first section comprises the first ring (carbons 9–14), the second carbon 16, the third carbons 19 and 20, the fourth carbon 23, and the fifth the last ring (carbons 25–30). For the peptide VTV, the SECCINT has a value of 3, that represents the first section with Carbon 4, the second with Carbons 6 and 7, and the last section with Carbon 10 (Fig. 1 (C)). Carbons belonging to groups C=O and COOC were not considered in the SECCINT, CH-INT or C-INT descriptors. The carbons that belong to hydrophobic sections are regarded as “external carbons” which are accounted for by the descriptor CARBEXT. The value of CARBEXT is 26 for the surfactant di-dodecyl dimethyl ammonium bromide, 20 for the 8-2-8, and 7 for the peptide VTV (Fig. 1 (A), Fig. 1 (B), and Fig. 1 (C)), respectively). The descriptors CH2-INT, CH-INT, C-INT, CH2-EXT, CH-EXT, and C-EXT only consider aliphatic atoms (i.e., carbons inside rings are not considered). Finally, hydrogens and atoms belonging to counter ions were not considered. All the 37 descriptors are shown in Table 1.

2.4. Model implementation and validation

An RF model was trained with the ST_{CMC} and the molecular descriptors of 691 surfactants (Supplementary Table S6) and 9 amino acids. The model was implemented in Python using the Random Forest Regressor of the library “sklearn”. All the parameters set for the model were left as default (Pedregosa et al., 2011). Five-fold cross-validation was used for evaluating the model performance on unknown molecules. This method is applied by randomly dividing all data into five groups with the same number of molecules and similar distribution of the ST_{CMC} . Then, the model is subsequently run by using each group as the validation set and the remaining as the training group. Six metrics were used for the model evaluation: correlation coefficient (r^2), mean absolute error (MAE), mean squared error (MSE), root-mean-square error (RMSE), index of agreement (d), and accuracy. The average of these metrics was computed and the standard deviation (StD) over the five validation groups. Importantly, the accuracy was, for the first time in the literature, considered in an ST_{CMC} predictive model to incorporate the systematic error into the evaluation of the model. This metric was defined as the number of predicted values whose difference with respect to the experimentally measured property was within ± 3 mN/m (based on the variation provided by Li et al. (Li et al., 2014) for the different measurements methodologies), divided by the total number of evaluated molecules.

2.5. ST_{CMC} estimation

The incorporation of different amino acids into the model was intended to improve the ST_{CMC} estimation. Amino acids fail to provide surface activity. However, this particular feature was used to

train the model in identifying molecular structures that are unlikely to induce significant changes in ST. Their inclusion in the model aided bridging the currently existing knowledge gap between conventional molecules (petrochemical surfactants) and peptides. This negative-case approach (i.e., molecules that fail to have the property estimated by the regression) for the adjustment of a regression model has been previously shown to improve property estimation in certain models (Cáceres et al., 2020). The model was applied to the prediction of the peptides ST_{CMC} . Small peptides can exhibit a CMC-type of behavior at various interfaces/surfaces (Infante and Moses, 1994; Zeng et al., 2009). In the case of long peptides, aggregation may not follow a micellar-like process (driven by interactions), but this agglomeration phenomenon is usually referred to as a critical aggregation concentration or CAC (for instance, for petroleum asphaltenes which are a solubility class family of compounds soluble in oil rather than a defined set of molecules, a Critical Nanoaggregate Concentration or CNAC is usually reported (Tian et al., 2013) (Lv et al., 2019). Regardless of the way through which surfactant molecules aggregate, the effect on the ST is the same: the ST decreases as the concentration increases but up to a certain concentration value after which further reduction is not observed. This minimum value is essential for the model independent of the supramolecular structures formed by the surfactant. Some of the ST_{CMC} values of the evaluated peptides were obtained from the literature, whereas others were measured in this study.

2.6. Ranking peptides according to predicted ST_{CMC}

Since short peptides are less prone to form secondary structures, permutations of peptides with less than six residues were considered here. Specifically, 17 amino acids were permuted in peptides with 2, 3, 4 and 5 residues. The tryptophan was excluded from the permutation because the double ring in its structure was not a characteristic found in the other surfactants of the database. Additionally, the amino acids aspartic acid and glutamic acid were disregarded because negatively charged molecules were not considered in the database for the model training. The reason for rejecting the inclusion of anionic and zwitterionic surfactants was to first set a baseline with simple molecules and then widen it in future studies. The descriptors of each peptide were automatically calculated by an algorithm in Python™ that summed up the contribution of each residue for each peptide descriptor. Based on the descriptors of each peptide, the RF model trained with conventional surfactants and amino acids was used to estimate the ST_{CMC} of the new peptides obtained after permutation.

3. Results and discussion

3.1. Data acquisition

Approximately 320 different literature sources for ST vs concentration data for nonionic and cationic surfactants were used to recover the ST_{CMC} values (Supplementary Table S6). Other sources of information might remain unexplored and could be considered further to increase the number of molecules in the database. Although the references were published in a long period of time (from 1967 to 2021), a lack of validity due to the use of obsolete elements was not evidenced by comparison with more recent publications. Recently, some authors also referred to studies older than 20 years for analyses that involve the ST_{CMC} (Rosen, 2004; Rosen, 2004). Many surfactant molecular structures were observed, including single head-chain, double chain, Gemini, and surfactants with different functional groups such as amine, hydroxyl, sulfur, aromatic, cyclic ester and carbon-carbon double bonds. Halo-

genated and silicon-based surfactants were not considered in the model because they have similar low ST_{CMC} values regardless of their molecular structure (Song et al., 2006). Therefore, they may negatively impact the accuracy and predictive power of the model. Because the low ST_{CMC} values of halogenated and silicon-based surfactants were absent in the database, the range of the ST_{CMC} varied from 21.92 mN/m to 55.2 mN/m. The ranges covered by the descriptors are shown in Table 1.

3.2. Model implementation and validation

The model was evaluated by fivefold cross-validation. All molecules were divided into five equally sized groups with a similar distribution of ST_{CMC} values, as shown in Fig. 2. This guaranteed that each validation included molecules in all the ST_{CMC} ranges. Fig. 2 also shows that most surfactants in the model have an ST_{CMC} between 27 and 42 mN/m. The most prevalent range of values was between 33 and 37 mN/m.

The model evaluation metrics are shown in Table 2. The average $r^2 = 0.55$ obtained in this model is low compared to previous studies, which estimated the CMC, a property closely related to the ST_{CMC} values higher than 0.90. A study from Gaudin et al. (Gaudin et al., 2018) for the ST_{CMC} of sugar-based surfactants and the application of the model over a validation set yielded an $r^2 = 0.78$ and MAE = 2.4 mN/m. The present model's MAE was 3.38 mN/m, as shown in Table 2. In other words, the metrics r^2 and MAE obtained in the current model indicate less accurate predictions than those obtained by the Gaudin work (Gaudin et al., 2018). These differences can be attributed to the wider variety of surfactants, not only sugar-based, covered here. This higher coverage implies that more data on each surfactant type is necessary for the model to learn the possibly different interfacial properties. A model trained with a higher number and types of molecules (i.e., cationic, neutral, linear, branched, with rings) allows to consider a higher number of structure-performance relationships in aspects such as the nature of the hydrophilic motif and which type has a stronger effect on the interfacial properties of molecules (Rosen, 2004). Considering into detail these relationships is important when the purpose of the model is to study the interfacial properties of molecules such as peptides because of their complex molecular structures, which present different types of hydrophobic and hydrophilic functional groups (e.g., linear carbons, aromatic rings, neutral amines, positively charged amines, hydroxyl groups, substituted rings, etc.). Table 2 also shows the accuracy, whose average for the validation groups was 0.77. This value indicates that almost 8 out of 10 tested molecules could be predicted within an uncertainty of ± 3 mN/m. The importance of the accuracy relies on the possibility of considering the experimental error in the model evaluation because interfacial properties are highly dependent on the purity of the sample components and the experimental approach used for the measurements (Takata et al., 2010; Ozeki and Ikeda, 1980; Staron et al., 2017). At this point, it is important to comment on the difference between systematic and random error. The systematic error considers the variations in the data that are inherent to the measurement techniques (i.e., type of measurement, calibration of the instruments, etc.), and that in this case, was set to ± 3 mN/m, as shown in section 2.4. On the other hand, the random error represents all the variations that arise as a consequence of the measurement process such as human error or lack of purity of the chemicals (Bowers et al., 2004). Despite its importance and to the best of the authors' knowledge, this is the first study that considers the model accuracy to complement the analysis of the interfacial properties prediction by comparison with the systematic error. The r^2 yielded a low value in our evaluations; however, this metric cannot consider the systematic error in the database values. Additionally, the random errors were also present

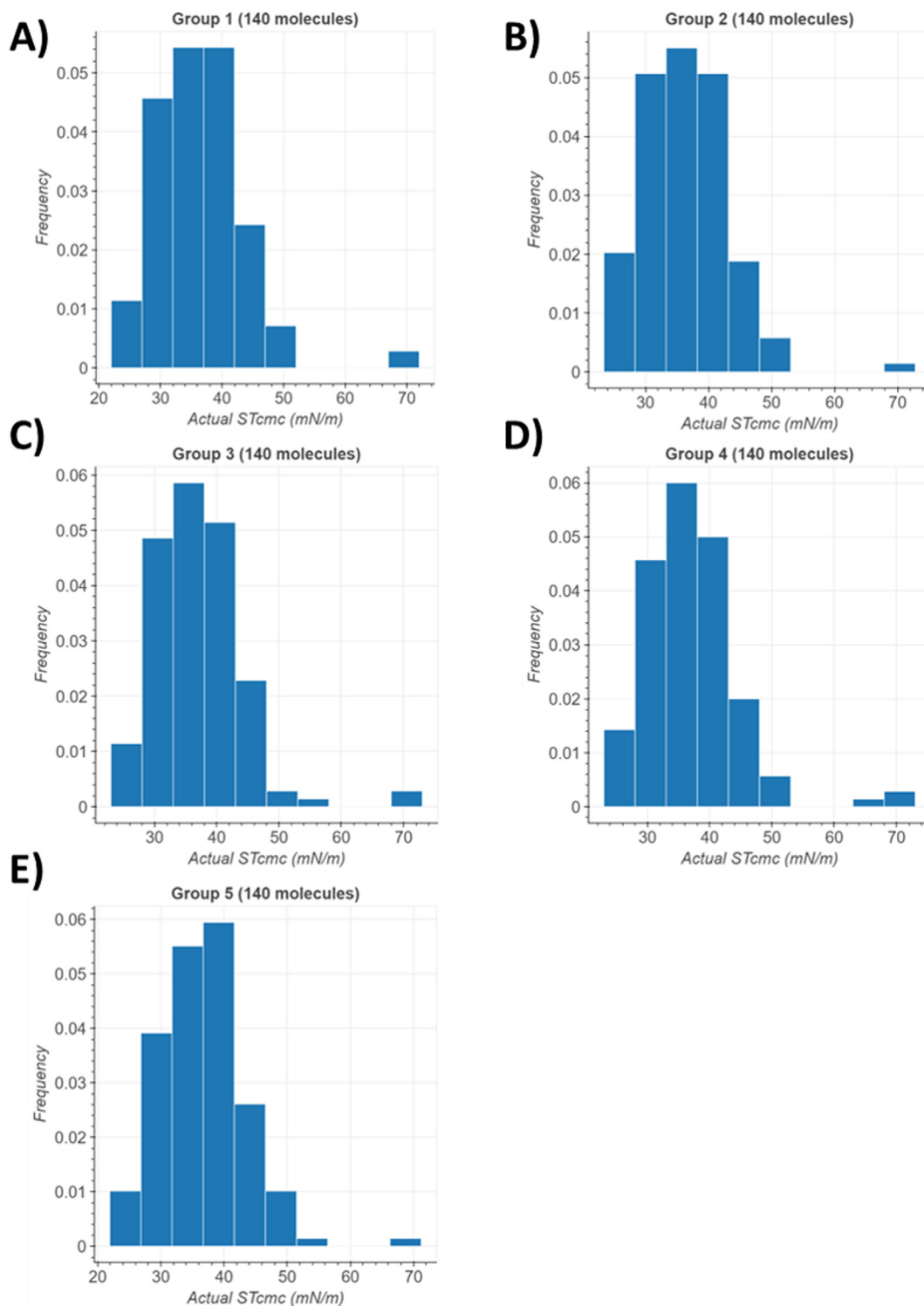


Fig. 2. ST_{CMC} distribution of the groups 1, 2, 3, 4 and 5 (A, B, C, and D, respectively), used in the model cross-validation. High similarity in the distribution and equal number of data points were the characteristics pursued in the data division.

in the model and reported in the literature (Li et al., 2014). Authors have discussed about the inconsistency between the ST_{CMC} values reported by various sources for the same molecules under identical experimental conditions. These differences were detected for some surfactants and eventually may reduce the model learning capacity and, therefore, its precision. For example, Bowers et al. (Łuczak et al., 2010) synthesized the surfactant 1-methyl-3-octylimidazolium chloride and measured its ST_{CMC} by the Wil-

helmy plate method at 298 K obtaining 37 mN/m. The same surfactant was evaluated by Łuczak et al. (Kuperkar et al., 2012) using the drop volume technique, and in their case the ST_{CMC} was 43.3 mN/m. Although the measurement techniques are fundamentally different in these studies (i.e., force vs drop shape), the experimental conditions were the same. Therefore, the differences related to the measurement method should not be significant and exceed 3 mN/m (Li et al., 2014). The Gemini cationic surfactant 16-2-16 was

Table 2

Model evaluation metrics for the five groups of the cross-validation. The average and standard deviation were also calculated.

Validation group	r^2	MAE	MSE	RMSE	d	a
1	0.63	3.04	17.3	4.16	0.86	0.79
2	0.56	3.39	20.0	4.47	0.84	0.74
3	0.59	3.42	21.6	4.65	0.86	0.78
4	0.57	3.35	19.9	4.46	0.85	0.76
5	0.39	3.69	34.7	5.89	0.74	0.80
Average	0.55	3.38	22.7	4.72	0.83	0.77
StD	0.08	0.21	6.14	0.60	0.04	0.02

synthesized by Kuperkar et al. (Aslam, 2019), and the ST_{CMC} measured by a Du Nouy tensiometer at 30 °C was 28.5 mN/m. Aslam evaluated the same surfactant using the platinum ring technique at 30 °C, and the resulting ST_{CMC} was 35 mN/m (Szymczyk et al., 2018). This is also the case of other cationic molecules in the database whose ST_{CMC} values vary significantly depending on the source. Li et al. (Li et al., 2014) hypothesized that these differences are due to the incomplete wetting occurring in the methods that use platinum (i.e., Wilhelmy plate and the du Nouy ring). For this reason, measurement methods that involve instruments made of metals have failed to provide the most accurate ST_{CMC} measurements for aqueous ionic surfactant solutions. A further data refinement in our database, only including reliable measurements, will be necessary in the future to reduce these random errors.

Priority must be given to measurement techniques that do not involve instruments made of metals. The pendant drop technique is the ideal candidate; however, this is the least common measurement technique employed for the surfactant solutions in the database. Exclusion of molecules from the database that were measured by a method different from the pendant drop is possible; however, this would imply removing a large body of data, a decision that probably would limit the learning of the model and thus a further r^2 decrease might be expected.

Concerning the differences between actual and predicted ST_{CMC} , these can be seen in Fig. 3 (A). The 45° line represents the location of properties whose real value was identical to the predicted one. In comparison, significant differences between real and anticipated ST_{CMC} are identified by longer distances between each point to the

45° line. Fig. 3 (A) shows that the model has a better performance predicting the ST_{CMC} for molecules in the range of 32–38 mN/m than for those in the extremes of the data dispersion (i.e., 27 and 42 mN/m). This behavior may be explained by the number of data incorporated into the model in each ST_{CMC} interval, as shown in Fig. 3 (B). The ranges 28 mN/m < ST_{CMC} less than 35 mN/m and 35 mN/m < ST_{CMC} less than 42 mN/m had the highest number of surfactants in the database (46 % of the total data), and also the highest accuracy of the model in comparison to the other data ranges where the number of surfactants is significantly limited. This direct proportionality between the ST_{CMC} data volume and the accuracy of the predictions indicates that a larger number of molecules allows broader learning of different structures and types of surfactants that produce ST_{CMC} values in a specific range. Fewer data in the other ranges generate incomplete learning and thus an inability of the model ST_{CMC} for correlation.

3.3. Peptides ST_{CMC} estimation

The model was used to estimate the ST_{CMC} of different peptides. Because the predicted molecules must have similar molecular characteristics to those of the database learning molecules, the model was trained with all the 691 surfactants. Additionally, nine amino acids were incorporated for training by measuring the largest ST reduction. Some ST reductions of the amino acids were obtained from other studies (i.e., alanine, phenylalanine, and leucine), while others were measured experimentally for the present work (i.e., lysine, arginine, glycine, tryptophan, methionine, and

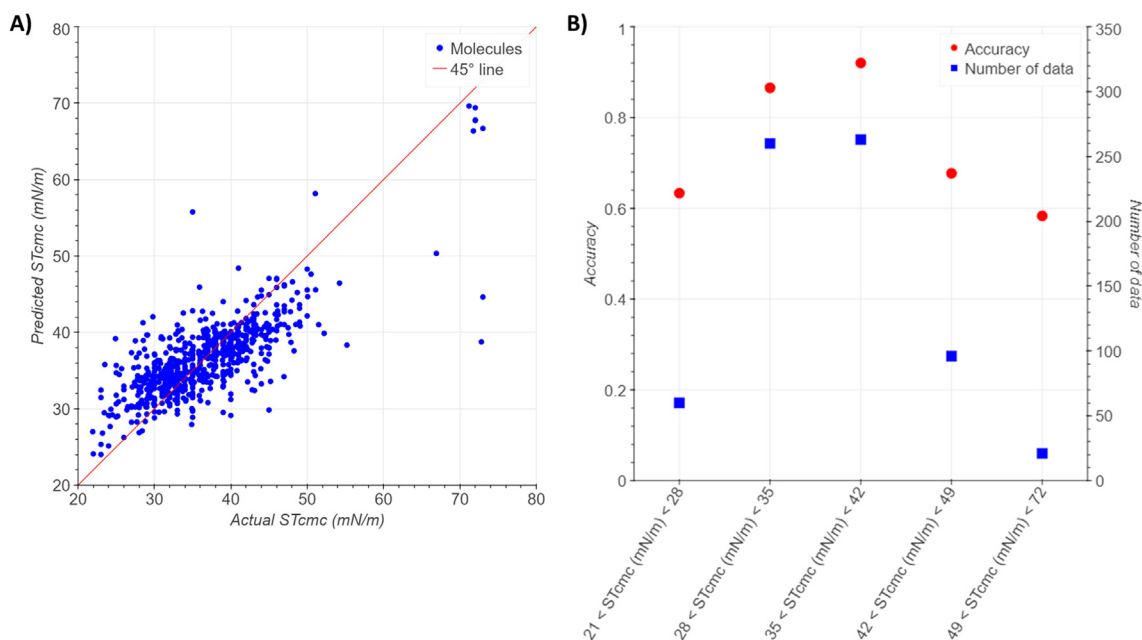


Fig. 3. (A) Similarity between the real and predicted ST_{CMC} from the database and the RF model. (B) Number of data for surfactants in each interval of ST_{CMC} .

histidine) (Chandra et al., 2013; Infante and Moses, 1994). The experimental results are shown in Table 3 for the amino acids L-lysine monohydrochloride, L-arginine, glycine, tryptophan, DL-methionine, and L-histidine. The ST_{CMC} of Tween 20 was also measured in this study as control to validate the ST_{CMC} measurement methodology employed.

Table 3 shows no reduction in the ST by adding the amino acids. Glycine cannot reduce the ST, as shown in the work of Chandra et al. (Chandra et al., 2013) where a minimum ST of 72.54 mN/m was measured, similar to the value obtained by us in the present study (73.00 mN/m). These results were expected because the size of the hydrophobic moieties in the amino acids is not big enough to induce the hydrophobic interactions necessary for interfacial activity (Ricardo et al., 2021). The values of the minimum ST of the amino acids evaluated in this study (Table 3) and others taken from the literature are shown in Table 4. The reliability of the surface tension measurement methodology employed here was validated by the ST_{CMC} measurement for Tween 20 (Fig. 4 (A)). The CMC of Tween 20 corresponds to the point in Fig. 4 (A) where the two straight lines intersect. The value of the ST at this point (ST_{CMC}) is 41 mN/m, which is the same value reported previously by another study using the ring method and the same temperature (Saito et al., 1995).

The RF model was used to predict the ST_{CMC} of 12 peptides. Ten of these peptides were taken from the literature, and the remaining two (RD and NL-19), were evaluated in this study by the same methodology used for the surfactant Tween 20. Table 5 provides the experimental and predicted ST_{CMC} values of the 12 peptides, and Fig. 4 (B) illustrates these data points.

Concerning peptides 4–10 of Table 5, although the experimental and predicted ST_{CMC} values ranged from 45 to 55 mN/m, some of the experimental values deviate from the predicted ones (Fig. 4 (B)). According to (Infante and Moses, 1994), the differences between ST_{CMC} values for peptides 4–10 are most likely due to the secondary structural features emerging from the amino acids interacting with themselves and the medium. It is comprehensible that the present RF model provides predictions solely based on 2D molecule structure descriptors. Thus, it fails to describe 3D molecule conformations such as secondary and tertiary structures. Different conformers of the same molecule may produce different

Table 4

Minimum IT of different amino acids in water solutions at 20 °C measured in this work and in other works.

Amino acid	Minimum IT value (mN/m)	Reference
Lysine (K)	72.8	This work
Arginine (R)	71.4	This work
Glycine (G)	73.0	This work
Methionine (M)	72.3	This work
Tryptophan (W)	73.0	This work
Histidine (H)	72.9	This work
Alanine (A)	72.3	(Chandra et al., 2013)
Phenylalanine (F)	70.1	(Chandra et al., 2013)
Leucine (L)	72.8	(Infante and Moses, 1994)

values for the interfacial tension (Edwards-Gayle and Hamley, 2017; James and Mandal, 2011). For this reason, the model tends to have precise ST_{CMC} predictions when peptides have fewer residues and by incorporating other parameters giving rise to secondary and tertiary structures may be needed to refine the model for longer peptides.

Peptides 11 and 12 are the longest from all the peptides in Table 5. They also have the largest differences between experimental and predicted ST_{CMC} , confirming the inability of the RF model to provide precise predictions for long peptides by only considering the 2D molecular descriptors. Some factors may cause this deviation. First, peptides 11 and 12 have aspartic acids (D) in their structure, a negatively charged amino acid. Although the net charge of the peptides was + 1, no descriptors used in model training accounted for the presence of negatively charged residues in the surfactants. For this reason, this factor could eventually cause differences in the predicted ST_{CMC} . Additionally, the same analysis of the presence of secondary structures discussed above applies for peptides 11 and 12. In fact, peptides with longer structures also exhibit the highest prediction errors. Secondary structures of peptides may cause hydrophobic moieties to not locate orthogonally to the air – water interface or even to prevent full contact with the hydrophobic phase. This lack of contact may eventually cause the molecule to lose the potential to strongly insert or attach to the air – water interface and the ST_{CMC} attained is not as small as predicted by RF model.

Table 3

ST vs concentration of six amino acids and the surfactant Tween 20 at 20 °C.

Molecule	Concentration (mg/L)	ST (mN/m) +/- 0.37	Molecule	Concentration (mg/L)	ST (mN/m) +/- 0.37
Lysine	6667	72.8	Methionine	6667	72.3
	222	73.0		222	72.6
	111	72.6		111	72.6
	74	73.2		74	72.5
	56	73.1		56	72.5
	44	73.3		44	72.7
Arginine	6667	71.5	Histidine	6667	73.3
	222	71.7		222	73.1
	111	71.5		111	73.1
	74	71.6		74	72.9
	56	71.9		56	73.1
	44	71.4		44	73.1
Glycine	6667	73.8	Tween 20	10,000	40.2
	222	73.0		3333	40.9
	111	73.0		200	46.2
	74	73.2		100	50.3
	56	73.2		67	51.0
	44	73.1		50	55.0
Tryptophan	400	72.9		40	56.1
	13	73.2			
	7	73.0			
	4	73.2			
	3	73.4			
	3	73.2			

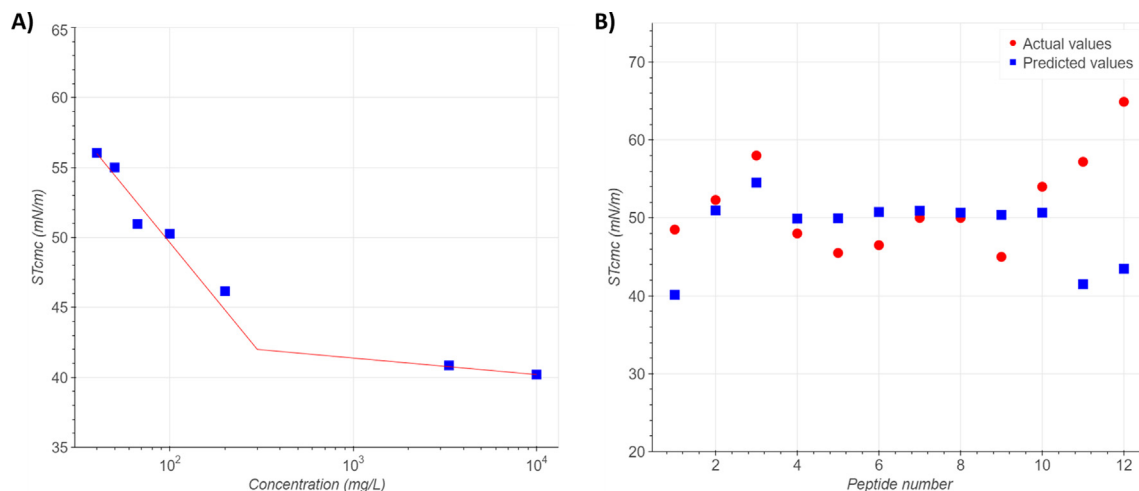


Fig. 4. (A) ST vs concentration plot of the surfactant Tween 20. The ST_{CMC} corresponds to the point where the two straight lines intersect. (B) Real and predicted ST_{CMC} values for the ten peptides studied in this work. The peptide number corresponds to the same numbers assigned in Table 5.

Table 5

Experimental and predicted ST_{CMC} of peptides by using the RF model trained with all the 691 molecules and 9 amino acids.

Number	Peptide	Real ST_{CMC} (mN/m)	Predicted ST_{CMC} (mN/m)	Difference ST_{CMC} (mN/m)	Reference
1	YF	48.50	40.13	8.37	(Zeng et al., 2009)
2	VTV	52.30	50.96	1.34	(Zeng et al., 2009)
3	QL	58.00	54.53	3.47	(Infante and Moses, 1994)
4	QLQL	48.00	49.91	1.91	(Infante and Moses, 1994)
5	QLQLQL	45.50	49.95	4.45	(Infante and Moses, 1994)
6	LQLQL	46.50	50.76	4.26	(Infante and Moses, 1994)
7	LLL	50.00	50.92	0.92	(Infante and Moses, 1994)
8	QQLLL	50.00	50.67	0.67	(Infante and Moses, 1994)
9	QLLL	45.00	50.38	5.38	(Infante and Moses, 1994)
10	QLLLQ	54.00	50.67	3.33	(Infante and Moses, 1994)
11	RTLFCRVGD (RD)	57.20	41.48	15.72	This work
12	IYKTPPIKDFGGFNFSQIL (NL-19)	65.00	43.47	21.43	This work

3.4. Ranking of less than 6-residue peptides by the predicted ST_{CMC} values

Permutations of 17 amino acids were developed for peptides with 5, 4, 3 and 2 residues to generate 1419857, 83521, 4913, and 289 combinations, respectively. For each peptide size (i.e., number of residues), the results were sorted by the predicted ST_{CMC} from the smallest to the highest value. Because of the large number of peptides, samples of the lowest, intermediate, and highest ST_{CMC} peptides were taken in the following way: 15 for the 2-residue, 50 for the 3-residue, 800 for the 4-residue, and 10,000 for the 5-residue ones. This sampling method was developed to include in the analysis a representative number of sequences of each section (i.e., smallest, intermediate, and highest ST_{CMC}). Additionally, in very close neighboring positions, the peptides had a high degree of similarity, therefore an analysis of different structures in each ST_{CMC} section would not have been possible. From these data, the appearance frequency of each amino acid was calculated as well as the average % of hydrophobicity for each permutation section (Supplementary Table S1). Finally, five peptides were selected from the lowest, intermediate, and highest ST_{CMC} ranges. The ST_{CMC} values of these peptides together with their molecular descriptors are shown in Supplementary Table S2, Table S3, Table S4 and Table S5 for the permutations 5, 4, 3, and 2; respectively.

The molecules with the lowest ST_{CMC} in Supplementary Table S2, Table S3, Table S4, and Table S5, seem to indicate that peptides fail to exhibit the molecular features required to attain

values of ST_{CMC} smaller than 44 mN/m, different to the majority of conventional surfactants (Fig. 2 (A), (B), (C), (D), and (E)). One of the main molecular features of conventional surfactants but absent in peptides is the presence of long hydrophobic carbon chains. These chains in peptides tend to be no longer than 4 carbons and never go above 8 or more carbons, a feature that is indispensable to confer a molecule with considerable interfacial properties (Rosen, 2004). This peptide constrain seems to be mitigated when the hydrophobic residues surpass 8 even if they are not all present in the same chain. However, the absence of chains with >4 carbons keeps the peptide from having hydrophobic domains that are large enough to increase the attraction and deposition of the peptide at the interfaces (Rosen, 2004; Scott et al., 2022). In consequence, the ST_{CMC} reduction induced by peptides is not as high as that provided by conventional surfactants.

The percentage of hydrophobicity (%) is defined as the percentage of hydrophobic residues divided by the total number of peptide residues. According to Supplementary Table S1, for all the permutations, a decrease in the peptide percentage of hydrophobicity produces higher ST_{CMC} values. In other words, the ability of the peptide to reduce the ST_{CMC} increases with the percentage of hydrophobicity. For the most ST_{CMC} reducing peptides, the percentage of hydrophobicity ranges between 50 % and 100 %. Conversely, the least ST_{CMC} reducing peptides have percentage of hydrophobicity below 50 % (Supplementary Table S1). High percentage of hydrophobicity for peptides with superior surface activity agree well with the experiments reported by Shigeri et al. (Shigeri et al., 2015), at least for long peptides, as shown in Table 6. The

Table 6

Experimental ST_{CMC} of peptides. Residues highlighted in red are hydrophobic, while those in black are hydrophilic (Shigeri et al., 2015). The percentage of hydrophobic residues is higher in peptides that attain a lower ST_{CMC} value.

Peptide	ST_{CMC} (mN/m)	% hydrophobic residues
IRPIPFIPRGGKT-NH2	72.01	69.2
FIGALLRPALKLLA-NH2	46.90	85.7
GLKEVAHSAKKFAKGFISGLTGS	72.01	56.5
LKGASKLIPHLLPSRQQ	72.01	52.9
FIGALLGPLLNLLK-NH2	38.30	85.7
IRPVFPFFPVHAKKVFLH	72.01	73.7

peptides with the lowest ST_{CMC} values are those with the highest ratio of hydrophobic residues (Table 6). This high hydrophobicity of good surfactants is necessary because these hydrophobic moieties provide higher chances of increased contact between the surfactant and the hydrophobic phase, therefore the insertion of the surfactant into the interface is energetically favored (Rosen, 2004). Additionally, the presence of larger hydrophobic domains is necessary in surfactants to reduce the total affinity of the molecule to the water (Rosen, 2004). If a surfactant lacks a large hydrophobic portion, it will not adsorb at the air – water or oil – water interfaces because it is more energetically favorable to remain in the water bulk (Rosen, 2004). The RF model was not used to estimate the ST_{CMC} of peptides in Table 6 because their length might probably induced the formation of secondary structural features, which the RF model fail to consider.

Supplementary Table S1 shows tendencies for the % of hydrophobicity of the 3 classifications in the four permutations; however, a number of exceptions can be observed when looking into particular peptides. For example, the peptides SGGSA (Supplementary Table S2), PGGC (Supplementary Table S3), GP and GG (Supplementary Table S5), have a percentage of hydrophobicity of 60 %, 75 %, 100 %, and 100 %, respectively, and they belong to the least ST_{CMC} reducing peptides zone. Although the prevailing modeling approach has focused on evaluating peptides in terms of their residues as a whole, in the case of short peptides, atomistic approximations seem to provide a better understanding in terms of the peptides interfacial activity. The peptides SGGSA, PGGC, GP and GC have been predicted to have poor interfacial properties if the molecular descriptor “external carbons” is considered, yielding values of 1, 0, 0 and 0, respectively. These external carbons are responsible for the strong adsorption tendency of surfactants at interfaces such as the oil – water or air – water (Rosen, 2004). If no or few external atoms are present, the interfacial activity of the molecule is almost negligible even if the % of hydrophobicity, from a residue point of view, is high.

The presence of hydrophilic and hydrophobic residues in the same peptide has been considered an essential feature due to their surface active properties (Ricardo et al., 2021). Supplementary Table S2 shows samples of the peptides with the highest ST_{CMC} reducing capabilities for the 5-residue permutation with the hydrophilic residues highlighted in red and the hydrophobic ones in black. All these peptides have the combination of hydrophilic and hydrophobic residues in the same molecule, that is in good agreement with the amphiphilic nature of surfactant peptides that other authors have identified (García-Moreno et al., 2020; García-Moreno et al., 2020; Li et al., 2020). However, the permutations of 4, 3 and 2 residues (Supplementary Table S3, Table S4 and Table S5, respectively) show that some of the peptides with the highest ST_{CMC} reducing capabilities have only combinations of hydrophobic residues such as the case of the peptides IFAI, MFLI, FLG, IFG, FII, FVL, IF, IV, LL, IA and ML. This rationale that is counterintuitive in light of the amphiphilic nature of surfactant peptides appears a valid alternative to design peptides with

interfacial activity and has been also put forward by a study of Scott et al. (Scott et al., 2022). According to their results, the application of molecular dynamics (MD) allowed to predict a high interfacial activity for the hydrophobic peptides GAMI, AGGI, AAMI, AAAL, IMLG, LAGA, GLAG and GIAA. The low ST_{CMC} values predicted for these totally hydrophobic peptides may break this rule of amphiphilic peptide surfactants; however, it appears to make sense from an atomistic rather than a residue view point. Considering that our RF model provides predictions for peptides based on molecular characteristics learned from conventional surfactants, for peptides with the highest ability to reduce the ST_{CMC} , it tends to maximize the number of external carbons by using residues such as isoleucine (I), leucine (L) and phenylalanine (F) (Fig. 5). Limiting issues of these highly hydrophobic peptides are their synthesis difficulty and their poor solubility in aqueous media, which complicates measuring the ST_{CMC} ; however, this is out of the model scope.

Supplementary Table S1 shows the frequencies in which each amino acid is present in all the evaluated peptides. The most ST_{CMC} reducing peptides of permutations 5, 4 and 3 have at least one I or L in their sequence and the presence of the amino acid F is mandatory (Supplementary Table S1). The same observations about the importance of the amino acids I and L in the most surface active peptides were obtained by Scott et al. (Scott et al., 2022) in the analysis of the ratio of molecules near the decane – water interface after MD simulations. Concerning the amino acid F, its importance in surface active peptides has been recognized by different studies that prove its ability to self-assemble and reduce the interfacial tension (Scott et al., 2016; Wychowanec et al., 2020). Although the amino acids I, L and F maximize the descriptor “external carbons”, whose importance is vital for surface active molecules, the most ST_{CMC} reducing peptides also include combinations with other amino acids such as methionine (M), alanine (A), glycine (G), valine (V) and lysine (K). In terms of the K, this is the most frequent hydrophilic residue in the most ST_{CMC} reducing peptides with an appearance frequency that approaches 60 % for the permutations of 4 and 5 residues (Fig. 5). The permutations of 3 and lower number of residues showed a decrease in the contents of K for peptides with the lowest ST_{CMC} . This low contents of K in 3-residue peptides with good surfactant properties was the reason by which Scott et al. (Scott et al., 2022) excluded this amino acid in the process of identifying a 3-residue templates for the design of 4-residue surfactant peptides. However, Fig. 5 suggests that the low frequency of this amino acid in 3-residue surfactant peptides with good interfacial properties may not be extrapolated for the rational design of longer peptides without altering the already superior interfacial properties. The main reason for this is that K can provide an important hydrophilic balance to long peptides without significantly altering the % of hydrophobicity. Regarding short peptides (i.e., less than 4 residues), the presence of K decreases the hydrophobicity of the molecule substantially and therefore the ST_{CMC} attained at the interface is high. Finally, despite that amino acids L, I and F provide the highest number of external carbons to peptides, which is in favor of interfacial properties, the peptide combinations with other nonpolar amino acids such as M, A, G and V are also predicted to be highly active interfacially by the RF model. These combinations are probably affected by other molecular descriptors different from the external carbons, whose contributions to the results are not easily identifiable but that the model considers important enough to avoid only combinations of L, I and F.

The RF model may not predict ST_{CMC} values that are sufficiently close to those obtained experimentally; however, the classification of the peptides according to the ST_{CMC} provides an initial approximation for subsequent in silico and experimental testing. For example, in the study of Scott et al. (Scott et al., 2022), the mea-

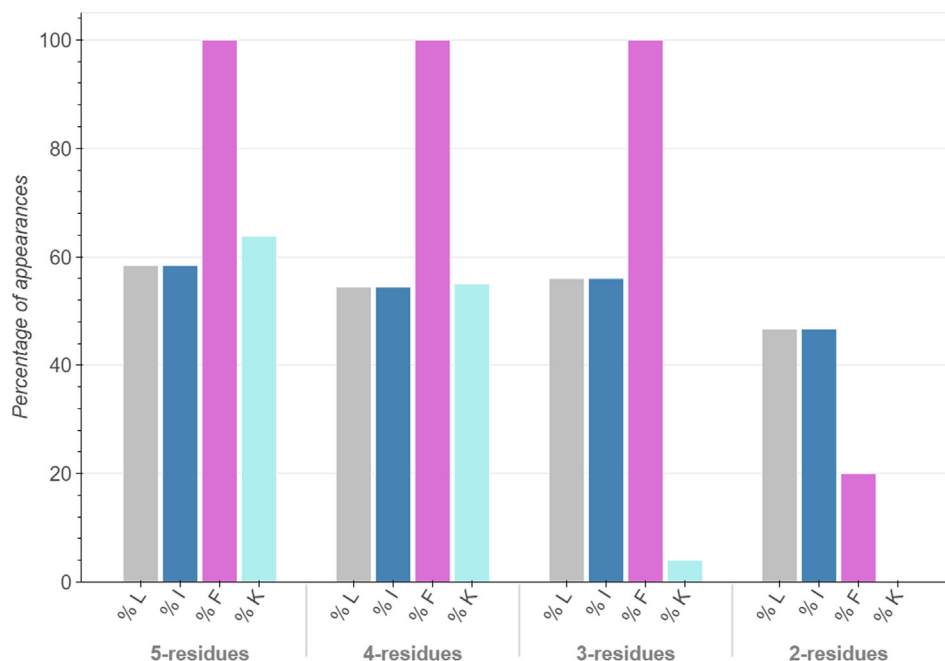


Fig. 5. Frequency of appearance of the amino acids L, I, F and K in the lowest-ranked ST_{CMC} peptides of the 5, 4, 3 and 2 residue peptides.

surement of the ST_{CMC} was developed for different 4-residues peptides as shown in Table 7. The absolute ST_{CMC} experimental values deviate from those estimated by the RF model; however, the classification of the peptides is almost identical. In Table 7, only the peptide LQCS, with 4 external carbons, is not in agreement with the ranking of the model, which was expected considering that for such low number of external carbons the model predicts a higher ST_{CMC} .

If we consider the results of the RF model and all the analysis reported previously, the peptides shown in Table 8 were the chosen for further experimental testing in the future as they exhibit the highest ST_{CMC} reduction.

According to the analysis, the amino acids I, L and F provide good surfactant characteristics which can be possibly extrapolated to long peptides. Also, the percentage of hydrophobicity must be above 60 % as shown in Table 6. These two rational design rules are the starting point for long surfactant peptides; however, more advances are necessary in terms of the amino acid sequences, motifs and interfacial mechanistic. Although some amino acids were identified as necessary for good surfactant peptides, the RF model fails to consider the order in that they are located within the peptide sequence. A detailed understanding of the underlying structure–function relationships may help gain insights into the phenomenon of the hydrophobic interface insertion. Additionally, a route to consider the secondary structure conformations in the model is mandatory for a more accurate estimation of the interfa-

Table 7

Peptides whose ST_{CMC} was evaluated experimentally by Scott et al. (Scott et al., 2022) and estimated by our RF model in the present study. The descriptor “external carbons” is also shown.

Sequence	External carbons	Experimental ST_{CMC} (mN/m)	Predicted ST_{CMC} (mN/m)
RRET	1	68.6	72.5
PTAL	6	66.4	43.1
LSQV	7	62.8	42.5
LQCS	4	60.3	48.5
HGII	8	51.7	42.0

Table 8

Peptides that are proposed for further experimental evaluation of ST_{CMC} . According to the RF model, these peptides are supposed to decrease the ST_{CMC} to the lowest value in comparison to the other peptides with the same number of residues.

Sequence	Number of residues
ILFQK	5
ILFK	4
LFG	3
IF	2

cial properties of long peptides as they have a considerable impact on the ST_{CMC} (Infante and Moses, 1994).

4. Conclusions

The most important molecular features of surface active molecules that can yield the lowest ST_{CMC} values can be identified using a AI model which correlates this property with molecular descriptors of conventional surfactants. This model can be applied to biomolecules such as peptides to identify amino acid combinations that maximize the ST_{CMC} reduction. A RF model was implemented with 691 surfactants and 9 amino acids to predict the ST_{CMC} of conventional and peptide surfactants. 37 molecular descriptors were extracted by the discrimination of the hydrophobic and hydrophilic sections of the molecule, the two main domains in surfactants. A fivefold cross validation process of the model was implemented and the metrics used for evaluation showed a low value of the coefficient of correlation in comparison to a prior ST_{CMC} study applied over sugar-based surfactants ($r^2 = 0.55$ vs $r^2 = 0.78$ (Gaudin et al., 2018)). A higher r^2 was difficult to attain due to data uncertainty, the lack of data in ST_{CMC} values closer to the limits (i.e., 27 and 42 mN/m), and random errors produced by the purity of synthesized surfactants and the inconsistency between the ST_{CMC} s provided by different authors under similar conditions. These differences suggest the need to refine the data in the data-

base by considering only molecules whose ST_{CMC} measurement does not involve the use of instruments made of metals (e.g., Wilhelmy plate), such as the pendant drop. Consequently, the addition of a large body of molecules is necessary to avoid the reduction in the model precision. To the best of our knowledge, for the first time, the accuracy was implemented in a ST_{CMC} estimation model to account for the systematic errors inherent to the measurement methodology in the analysis of the model.

The model was used to predict the ST_{CMC} of 10 peptides taken from the literature and 2 peptides measured experimentally in this work. The model was successful in predicting that peptides fail to have molecular characteristics to attain ST_{CMC} values smaller than 38 mN/m. The ability of the model to predict ST_{CMC} values similar to the real property decreased as the size of the peptide increased. These differences suggest the inability of the model to consider the possibility of forming 3D molecular conformations such as secondary and tertiary structures. For this reason, the improvement of the prediction for large peptides must include factors that incorporate characteristics of secondary and tertiary structures of peptides, and the pattern of distribution of the amino acids in the peptide that is ignored by the RF model.

Finally, the model was applied over a permutation of peptides with less than 6 residues. The molecular characteristics of the most and the least ST_{CMC} reducing peptides were in good agreement with those observed for conventional surfactants, in which a number of external carbons higher than 8 is necessary to confer interfacial properties (Rosen, 2004). The hydrophobic moieties of the molecules were maximized by the RF model in the low ST_{CMC} peptides and therefore the presence of the amino acids L, I and F was mandatory. This predominance of the amino acids L and I was also obtained in other study by means of MD and experiments (Scott et al., 2022). Peptides with no external carbons failed to substantially reduce the IT of water according to the predictions of the model. Other molecular characteristics additional to the high hydrophobicity of the molecules may have affected the model ability to predict the ST_{CMC} for the peptides because the residue order was not strictly based on the size of the hydrophobic moieties. According to the analysis of the molecular characteristics for the most promising ST_{CMC} reducing peptides, the peptides ILFQK, ILFK, LFG and IF, were proposed in each permutation size for future experimental evaluation.

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CRediT authorship contribution statement

Fabián Ricardo: Investigation, Data curation, Writing – original draft. **Paola Ruiz-Puentes:** Software, Visualization. **Luis H. Reyes:** Conceptualization, Methodology, Writing – review & editing. **Juan C. Cruz:** Conceptualization, Methodology, Writing – review & editing. **Oscar Alvarez:** Conceptualization, Methodology, Writing – review & editing. **Diego Pradilla:** Conceptualization, Methodology, Writing – review & editing.

Data availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Percentage of peptides that have each of the listed amino acids in the lowest, intermediate, and highest ST_{CMC} values regions according to the number of residues (i.e., 5, 4, 3 and 2). Five-residue peptides with the lowest, intermediate, and highest predicted ST_{CMC} values and their respective descriptors. Four-residue peptides with the lowest, intermediate, and highest predicted ST_{CMC} values and their respective descriptors. Three-residue peptides with the lowest, intermediate, and highest predicted ST_{CMC} values and their respective descriptors. Two-residue peptides with the lowest, intermediate, and highest predicted ST_{CMC} values and their respective descriptors. The database of conventional surfactants used for the model training and evaluation. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2022.118208>.

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